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Validation of seasonal mean radiant temperature simulations in hot arid urban climates

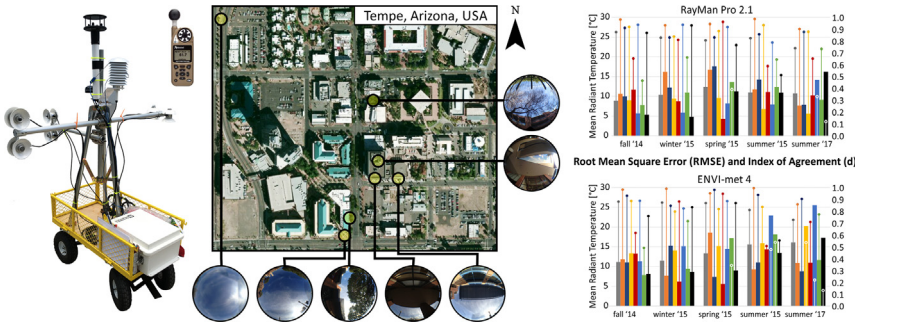
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HIGHLIGHTS

- Ray Man and ENVI-met fail to accurately model T_{MRT} under extreme heat.
- Seasonal T_{MRT} errors exceed suggested accuracy of ±5 °C for heat stress studies.
- Models show seasonal swing in RMSE with increased errors under high solar radiation.
- Ray Man struggled most with engineered enclosed spaces and complex urban forms.
- ENVI-met struggled with engineered shade and did not resolve spatial variability.

GRAPHICAL ABSTRACT



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ABSTRACT

We validated seasonal RayMan and ENVI-met mean radiant temperature (T_{MRT}) simulations to assess model performance in a sensitivity analysis from cold to extremely hot conditions. Human-biometeorological validation data were collected in Tempe, Arizona via transects during five field campaigns between 2014 and 2017. Transects were conducted across seven locations in two to three-hour intervals from 6:00 to 23:00 LST with a Kestrel meter and thermal camera (2014–2015) and the mobile instrument platform MaRTy (2017). Observations across diverse urban forms, sky view factors, and seasons covered a wide range of solar radiation regimes from a minimum T_{MRT} of 8.7 °C to a maximum of 84.9 °C. Both models produced large simulation errors across regimes with RMSE ranging from 8 °C to 12 °C (RayMan) and 11.2 °C to 16.1 °C (ENVI-met), exceeding a suggested T_{MRT} accuracy of ±5 °C for heat stress studies. RayMan model errors were largest for engineered enclosed spaces, complex urban forms, and extreme heat conditions. ENVI-met was unable to resolve intra-domain spatial variability of T_{MRT} and exhibited large errors with RMSE up to 25.5 °C for engineered shade. Both models failed to accurately simulate T_{MRT} for hot conditions. Errors varied seasonally with overestimated T_{MRT} in the summer and underestimated T_{MRT} in the winter and shoulder seasons. Results demonstrate that models should not be used under micrometeorological or morphological extremes without in-situ validation to quantify errors and assess directional bias due to model limitations.
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1. Introduction

While the world continues to urbanize with a projected 68% of the population living in cities by 2050 (UN, 2019), an anticipated increase in the length, frequency, and intensity of extreme heat events threatens the health and livelihood of urban populations (Jones et al., 2015). The combined effects of urbanization and global-scale climatic change on local and microscale urban climate will likely lead to increased human exposure to heat, driving thermal discomfort and raising the risk of heat-related illness and mortality, especially for vulnerable populations.

In the advent of a warming world, a growing body of literature has investigated the impact of extreme heat on urban populations using a variety of thermal metrics, from air temperature (T_a) and surface temperature (T_s) to radiant temperature and complex indices that combine several atmospheric variables. Under hot, dry conditions, human thermal exposure is mainly driven by the radiative environment (Lee et al., 2013; Mayer et al., 2008), and mean radiant temperature (T_{MRT}) has emerged as the most important atmospheric variable for outdoor thermal comfort (Chen and Ng, 2012; Johansson et al., 2014; Middel et al., 2016). T_{MRT} is a synthetic parameter that integrates short- and longwave radiation flux densities a human body is exposed to into a single value (ISO, 1998). This parameter is the basis for many human-biometeorological indices, such as Physiologically Equivalent Temperature PET (Mayer and Höppe, 1987) and the Universal Thermal Comfort Index UTCI (Bröde et al., 2012).

Numerous models have been developed to simulate thermal conditions in outdoor spaces at people-relevant scales – from neighborhoods to individual features (e.g., trees). Models mainly differ in their spatial extent (single location vs. 2D or 3D grid), spatial and temporal resolution, computational intensity, input requirements, the type of output heat metrics (e.g., T_a, T_{MRT}, thermal comfort indices), and calculation of atmospheric conditions (empirical vs. physics-based). Models have been applied to assess thermal impacts of urban form and design scenarios (Emmanuel et al., 2007; Heris et al., 2020; Middel et al., 2014; Yahia and Johansson, 2013), the effectiveness of heat mitigation strategies such as increased tree canopy cover and cool roofs (Aminipouri et al., 2019a, 2019b; Crank et al., 2018; Middel et al., 2015; Zölch et al., 2016), human outdoor thermal comfort (Lee et al., 2016; Lin and Matzarakis, 2008; Morakinyo et al., 2017), and the relationship between heat and mortality (Thorsson et al., 2014).

RayMan (Matzarakis et al., 2010) and ENVI-met (Bruse and Fleer, 1998) have most frequently been used to assess thermal conditions in complex urban environments at the microscale. A Scopus search in June 2020 for the past 20 years yielded 87 RayMan and 423 ENVI-met studies. While RayMan was specifically designed to model radiation flux densities and T_{MRT} for single points in space, ENVI-met is a numerical model that simulates a full suite of atmospheric variables in a gridded domain. Both models were developed in a temperate climate (Germany), but have been employed in a wide array of urban and climatic contexts. Although these models have previously been used in hot, arid environments (e.g., Chow and Brazel, 2012; Hedquist and Brazel, 2014; Mahmoud, 2011), their ability to accurately simulate daytime T_{MRT} in those regions has not been assessed systematically by season. Additionally, with recent trends in urban climate scholarship to investigate human thermal exposure under extreme heat, the need emerges to evaluate if the models produce accurate results for extreme conditions.

Recognizing the impact future urbanization and climate will have on human vulnerability to extreme conditions, the goal of this study is to validate seasonal RayMan and ENVI-met T_{MRT} simulations for heterogeneous urban forms using in-situ observations in Tempe, Arizona. Model performance is assessed in a sensitivity analysis from cold to extremely hot conditions across a wide range of solar radiation regimes to determine: Are RayMan and ENVI-met heat-ready?

2. Methods

2.1. Study area

This study was conducted in Tempe (33°25′28.6″N, 111°56′18.6″W), a city in the East Valley of the Phoenix metropolitan area in Maricopa county, Arizona, USA (Fig. 1). Located in the Sonoran Desert of the American Southwest, the 192,000 Tempe residents experience a subtropical arid climate (Köppen Climate Classification subtype Bwh, tropical and subtropical desert climate) with daytime maximum T_a exceeding 40 °C for most of June, July, and August (period of record: 1953–2013). While maximum summer T_a can peak close to 50 °C, overnight lows often are between 27 °C and 29 °C (Western Regional Climate Center, 2020). Mean annual rainfall is relatively low (219 mm, period of record: 1953–2013) with 40% of the annual total occurring in the summer months due to the North American Monsoon.

Most of the City of Tempe can be classified as open low-rise Local Climate Zone (Stewart and Oke, 2012) with detached one- to two-story single-family homes and lower density development patterns. The study area in Downtown Tempe is denser with three- to four-story office and commercial buildings (Fig. 1) and mixed-use, high-rise apartments currently under construction, transitioning from an open midrise to open high-rise zone. The Mill Avenue District is a shopping and entertainment area north of Arizona State University's Tempe campus. Streetscapes are lined with stores, restaurants, and businesses; sidewalks are canopied with street trees; and narrow pedestrian corridors connect small courtyards and plazas. In contrast, the Mill Avenue District is bordered by an arterial to the south (University Drive) and a collector road to the west (Ash Avenue) with low-rise buildings, parking lots, and little shade.

Building on human-biometeorological observations from a previous study on outdoor thermal comfort in Tempe's Mill Avenue district (Crewe et al., 2016), seven sites were chosen for continued seasonal monitoring to validate T_{MRT} simulations in the area (Fig. 1). The sites provide a good tradeoff between limited transect walking distance and maximum spatial coverage to validate modeled intra-neighborhood T_{MRT} distribution. The locations cover a range of Sky View Factors (amount of visible sky in the upper hemisphere), varying amounts of vegetation, and different canyon orientations (Table 1).

2.2. Field observations

We conducted five clear-sky field campaigns to collect seasonal human-biometeorological data in the study area for model validation: on October 24, 2014 (fall), February 15, 2015 (winter), March 23, 2015 (spring), June 20, 2015 (summer), and June 21, 2017 (summer). During the first four campaigns (2014–2015), we used a Kestrel 4400 heat stress tracker to collect T_a, relative humidity (RH), wind speed ($_v$), and globe temperature (T_g) at 1.1-m height – the center of gravity of a standing human (ISO, 1998) – on transects across all sites (Supp. Fig. 1a). Observations were performed in three-hour intervals from 06:00 to 21:00 and again at 23:00 Local Standard Time (LST). Each human-biometeorological transect took an hour to complete, with seven-min stops at every site to account for the globe thermometer's slow response time (Middel et al., 2016). During the summer and winter field campaigns, T_s was collected at each site using a FLIR i3 thermal imaging infrared camera facing in all cardinal directions and downwards (Supp. Fig. 1b). Each site was documented through hemispherical 180° photos taken at 1.1 m height with a Canon EOS 6D and Canon EF 8–15-mm f/4 Fisheye USM Ultra-Wide Zoom lens.

With the availability of new equipment, we completed an additional field work campaign on June 21, 2017 using a Kestrel 4400 heat stress tracker and the mobile sensor platform *MaRTy* (Middel et al., 2020; Middel and Krayenhoff, 2019). *MaRTy* observes georeferenced 6-directional longwave (L) and shortwave (K) radiation flux densities with three Hukseflux 4-Component Net Radiometers, T_a, T_s, v, and RH

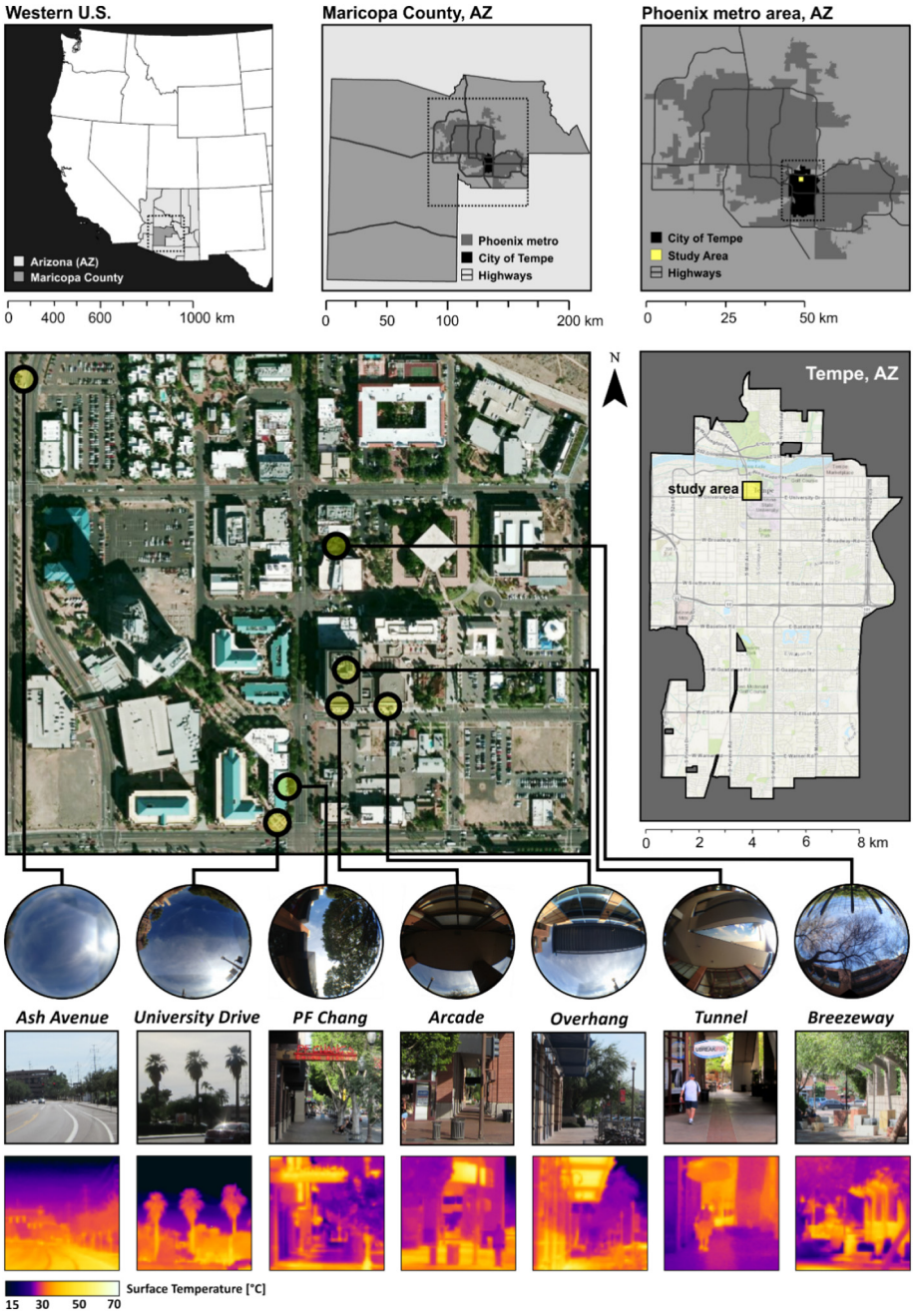


Fig. 1. Aerial view of study area in Downtown Tempe, Arizona, USA with hemispherical fisheye views for each stop along a human-biometeorological transect, corresponding photos, and thermal images (October 24, 2014, 15:00 Local Standard Time).

Table 1
Field campaign site characteristics: albedo of ground surface, surrounding vegetation fraction, Sky View Factor (SVF), canyon orientation, and site description. SVF was calculated from fisheye photos (Middel et al., 2018); surrounding vegetation fraction on a sphere was calculated from image cubes (Middel et al., 2019). For corresponding fisheye photos see Supp. Fig. 2.

Site	Albedo	Vegetation fraction (%)	Sky view factor	Canyon orientation	Site description
Ash Avenue	0.13	5	0.948	N-S	Exposed N-S street lined by scattered trees; site on brick west sidewalk
Arcade	0.12	9	0.050	E-W	Brick sidewalk covered by brick overhang-columned arcade structure on N side of E-W street
Breezeway	0.22	39	0.436 (spring, summer, fall) 0.123 (winter)	E-W	E-W walkthrough from small plaza to the east to main N-S street to the west; buildings to north and south of walkthrough; colored brick surface; seasonal deciduous arbor tree canopy over walkthrough
Tunnel	0.14	3	0.030	E-W	E-W walkthrough from main N-S street to open plaza; building tunnel
Overhang	0.12	11	0.213	E-W	Thin metal shade overhang structure over brick sidewalk on N side of E-W street
Mill Avenue	0.15	18	0.190	N-S	West sidewalk of Mill Avenue with large tree canopy year-round for shade
University Dr	0.23	17	0.783	N-W corner	Open NW street corner brick/cement concrete sidewalk with buildings to N and W

in 2-s intervals at pedestrian height (Supp. Fig. 1c, Table 2). Human-biometeorological transects were performed every two hours from 8:00 to 14:00 LST, at 17:00 LST, and after sunset at 20:00 LST, following the same traverse protocol as in previous campaigns.

2.3. Data processing

All T_a observations were linearly time-detrended to the full hour using Sky Harbor airport data. Kestrel observed T_g [°C] was converted to T_{MRT} [°C] using the on-site measurements of T_a [°C] and v [ms^{-1}], assuming a globe emissivity ε of 0.95 and an adjusted globe diameter D of 150 mm (ISO, 1998; Thorsson et al., 2007):

$$T_{MRT} = \left[(T_g + 273.15)^4 + \frac{1.1 \times 10^8 v^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{1/4} - 273.15 \quad (1)$$

T_s was extracted from the thermal images using the FLIR Tools IR software, and an areal average T_s was calculated for each site and hour (Häb et al., 2014). *MaRTy* observations were cleaned, i.e. the first minute of each transect stop was discarded to account for sensor lag (Häb et al., 2015), and the remaining transect observations at each site were averaged. T_{MRT} was calculated from recorded 6-directional radiation flux densities K_i and L_i [Wm^{-2}] using absorption coefficients $a_k = 0.07$ (shortwave) and $a_l = 0.97$ (longwave), the Stefan-Boltzmann constant σ, and weighting factors $W_i = 0.06$ (up/down) and $W_i = 0.22$ (cardinal directions) for a standing person (Höppe, 1992; ISO, 1998):

$$T_{MRT} = \left[\sum_{i=1}^{6} \frac{W_i (a_k K_i + a_l L_i)}{a_l \sigma} \right]^{\frac{1}{4}} - 273.15 \quad (2)$$

2.4. RayMan simulations

The RayMan model calculates longwave and shortwave radiation flux densities, T_{MRT}, SVF, sun duration, and thermal comfort indices for a single location based on meteorological inputs and hemispherical photos (Matzarakis et al., 2007, 2010). At a minimum, RayMan requires T_a, v, vapor pressure or *RH*, cloud cover, latitude and longitude, time of day, and day of year to estimate T_{MRT} from a fisheye photo that captures the surrounding 3D urban environment in 2D. Optional input parameters include the average albedo (α) and T_s of surrounding surfaces, the Bowen ratio (ß), and global radiation (G). Additionally, the user can specify personal characteristics (height, weight, age, gender), clothing, and activity to tailor thermal comfort outputs to a specific demographic.

We ran 70 RayMan Pro 2.1 simulations using the fisheye photos collected on-site. For each of the seven transect stops, we simulated two stages of meteorological forcing from the seasonal field campaign data (Supp. Table 1): 1a) Kestrel T_a, v, and *RH*; 1b) Kestrel T_a, v, *RH*, and FLIR-derived T_s; 2a) *MaRTy* T_a, v, and *RH*; 2b) *MaRTy* T_a, v, *RH*, and T_s from the downward facing pyrgeometer.

2.5. ENVI-met simulations

ENVI-met (Bruse and Fleer, 1998) is a non-hydrostatic Computational Fluid Dynamics (CFD) model used to simulate microscale interactions between surfaces, vegetation, and the atmosphere in the urban canopy layer. We used ENVI-met Science version 4.3 released in the winter of 2017/2018 to model human-biometeorological conditions in the Mill Avenue District for the five field campaign days and test the model's ability to simulate T_{MRT} across space, times of day, and seasons. The input domain size was 190 × 195 × 25 grid cells with 20 nesting grids (Supp. Fig. 3) at a grid cell size of Δx = Δy = Δz = 4 m and 20% telescoping after z = 40 m. The domain was designed larger than the study area to minimize edge effects. Building wall materials were set to moderate insulation (building plan fraction 22.6%), and plants were a mix of evergreen, deciduous, and palm trees with little grass (vegetation plan area fraction 9.1%). Soil conditions matched values reported by Middel et al. (2014). We ran ENVI-met for a 24-hour period starting at 03:00 LST with a constant time step of (2 s) and model output every 60 min. The model was forced with hourly T_a, *RH*, and all-day prevailing wind speed and direction (fixed) from the nearby Sky Harbor International Airport (ENVI-met "simple forcing", standard TKE model) (Fig. 2).

2.6. Model validation

To quantify the model accuracy of RayMan in simulating T_{MRT}, we fit linear regression models to the observed field work data and report the coefficient of determination (R^2) to investigate how much variance can be explained in the observed (O) data by the model-predicted (P) data. Further testing the reliability of results, we report five performance statistics: the mean bias error (MBE), root mean square error (RMSE), its systematic and unsystematic components $RMSE_s$ and $RMSE_u$, mean absolute error (MAE), and Wilmott's index of agreement (d) (Willmott, 1981). The index of agreement (d) evaluates the error between observed and modeled data as a standardized measure between 0 and 1 (a value of 1 indicates perfect agreement):

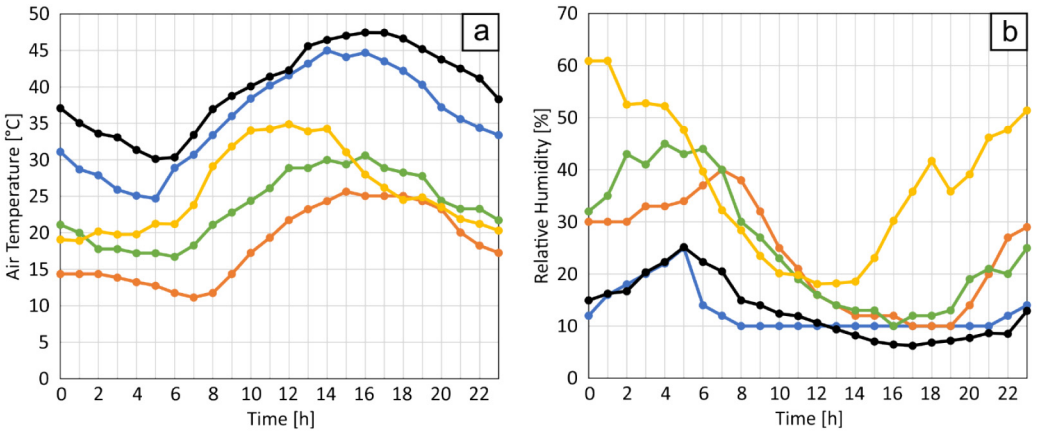
$$d = 1 - \left[\frac{\sum_{i=1}^{N} (P_i - O_i)^2}{\sum_{i=1}^{N} \left(|P'_i| - |O'_i| \right)^2} \right]$$

where $P'_i = P_i - \overline{O}$, $O'_i = O_i - \overline{O}$, and N is the number of cases.

To assess the performance of ENVI-met, we first evaluate the model's ability to simulate T_a, which is governed by the urban canopy layer turbulent transport, before evaluating simulated T_{MRT} using the same performance statistics.

Table 2
Summary of sensor specifications and accuracies.

Sensor	Variable	Range	Accuracy
Kestrel 4400 heat stress tracker	Air temperature	−10.0 to 55.0 °C	0.5 °C
	Globe temperature	−10.0 to 55.0 °C	1.4 °C
	Wind speed	0.6 to 40.0 ms^{-1}	0.2 ms^{-1}
	Relative humidity	0 to 100%	3.00%
FLIR i3	Surface temperature	−20 to 250 °C	± 2 °C
HC2S3 Rotronic HygroClip2	Air temperature	−50 to 100 °C	± 0.1 °C
T/RH Probe	Relative humidity	0 to 100%	±0.8 RH
Gill 2D WindSonic	Wind speed	0 to 60 ms^{-1}	± 2% @ 12 ms^{-1}
GPS16X Garmin GPS	Lat/Lon	−90 to 90°	<3 m
		−180 to 180°	
3 Hukseflux 4-component net radiometers	SW radiation	285 to 3000 × 10^{-9} m	±10%
	LW radiation	4.5 to 40 × 10^{-6} m	±10%



	2014-10-24	2015-02-18	2015-03-23	2015-06-20	2017-06-21
start time	3:00 AM	3:00 AM	3:00 AM	3:00 AM	3:00 AM
hours of simulation	24	24	24	24	24
wind speed at 10 m [m s^{-1}]	1.5	2.6	3.1	1.0	1.0
wind direction [°]	270	270	270	270	270
roughness length [m]	0.01	0.01	0.01	0.01	0.01

Fig. 2. Summary of forcing data and parameters from Sky Harbor International Airport for seasonal ENVI-met simulations: a) hourly air temperature; b) hourly relative humidity; tables with predominant wind speed/direction from Sky Harbor International Airport and other simulation parameters.

3. Results

3.1. Seasonal human-biometeorological field observations

T_{MRT} varies significantly between locations throughout the day for all seasons (Fig. 3). Variability is mainly driven by the availability or absence of shade, which modulates the contributions of incoming shortwave radiation to T_{MRT}. Open locations (Ash Avenue and University Drive) exhibit consistently higher T_{MRT} than T_a except in the early morning hours. Differences are most pronounced at 12:00 LST in the summer (23.5 °C–43.3 °C) and at 9:00 LST in the winter (36.2 °C–40.7 °C) and shoulder seasons (18.8 °C–50.7 °C). In contrast, the tunnel location is exposed to minimal diurnal incoming shortwave radiation that peaks at low sun angles. While T_{MRT} observed in the morning is subject to lateral shortwave radiation in winter and spring, T_{MRT} is close to T_a in the afternoon all year round when the enclosing building blocks the sun completely. In the summer, T_{MRT} is slightly below T_a for most parts of the day (0.6 °C – 8.1 °C lower). The two east-west locations (Overhang and Arcade) are sun-exposed from late morning into the afternoon during fall (max T_{MRT} = 76.5 °C), winter (max T_{MRT} = 43.6 °C), and spring (max T_{MRT} = 66.0 °C), because they are on the sidewalk to the north and not shaded by tall buildings to the south. In the summer, the Arcade provides shade all day due to the high sun elevation (max T_{MRT} = 66.0 °C), while the overhang site is sun-exposed, because the overhang is on the south side of the building. The North-South street canyon site (Mill Avenue), which receives reduced direct shortwave radiation in the morning because of a tree to the east, is briefly sun-exposed around noon, and fully shaded by a building for the rest of the day. T_{MRT} peaks at 46.1 °C in the spring, 61.4 °C in the summer, 50.6 °C in the fall, and 32.0 °C in the winter. The vegetated breezeway exhibits a random T_{MRT} pattern. A building and a deciduous tree to the south provide shade most of the day, but the site is occasionally hit by direct sunlight, penetrating the canopy.

Comparing observations across summer field work campaigns, T_{MRT} at the University Drive site at 12:00 LST on June 20, 2015 is 17 °C higher than on June 21, 2017 despite similar sun elevation angles and diurnal T_a profiles (max Ta = 44.1 °C vs. 47.4 °C). This is attributed to the difference in sensor setup and wind conditions (see Discussion).

3.2. RayMan validation

To examine our RayMan simulations, we used multiple error metrics to assess the validity of the mode. For all seasons, times of day, and combined receptors (N = 238), RMSE is 10.7 °C with MBE = −0.5 °C and d = 0.88. Across all sites, times of day, and seasons (N = 49), RayMan simulated T_{MRT} exhibits an RMSE of 8.0° to 12.4 °C (Fig. 4, Supp. Table 2). The error remains high irrespective of the instrumentation used for in-situ observations (Kestrel vs. MaRTy). MBE is positive in the summer but negative for winter and shoulder seasons, i.e. RayMan overestimates T_{MRT} in the summer and underestimates T_{MRT} otherwise. Adding T_s as a forcing parameter yields similar performance metrics overall with RMSE ranging from 9.1–14.1 °C (Supp. Table 3). The unsystematic error is always larger than the systematic error, which reflects the heterogeneity of the sites in terms of urban fabric and horizon limitations.

RayMan performs best for the open site Ash Avenue (SVF = 0.948) with an index of agreement of 0.98 in the fall and 0.90–0.99 in the summer, but yields an RMSE of 16.2 °C in the winter and 16.7 °C in the spring despite an acceptable index of agreement (d = 0.94 and 0.93, respectively). RayMan underestimates T_{MRT} consistently for open sites independent of season with negative MBE values between −7.4 °C (June 2017) and −14.9 °C (February 2015). Consequently, the systematic error for open sites is higher than the unsystematic error.

The tunnel reveals a seasonal influence on simulation performance; RMSE is lowest in the winter (4.9 °C), increases with higher T_a, and peaks in June 2017 with 16.2 °C. MBE is positive for all seasons, i.e. RayMan consistently overpredicts T_{MRT} in the tunnel, and MBE mirrors the RMSE trend with the lowest value in the winter (1.0 °C), an increase during shoulder seasons (2.7 °C and 7.7 °C), and a maximum error of 14.1 °C in the summer. Results are improved with T_s as an additional forcing parameter, but still yield errors over 10.0 °C for June 2017. The index of agreement shows no significant agreement between predicted and observed summer values for the tunnel (d = 0.51 in June 2015, d = 0.12 in June 2017). These findings suggest that RayMan is not able to accurately simulate engineered enclosed outdoor spaces.

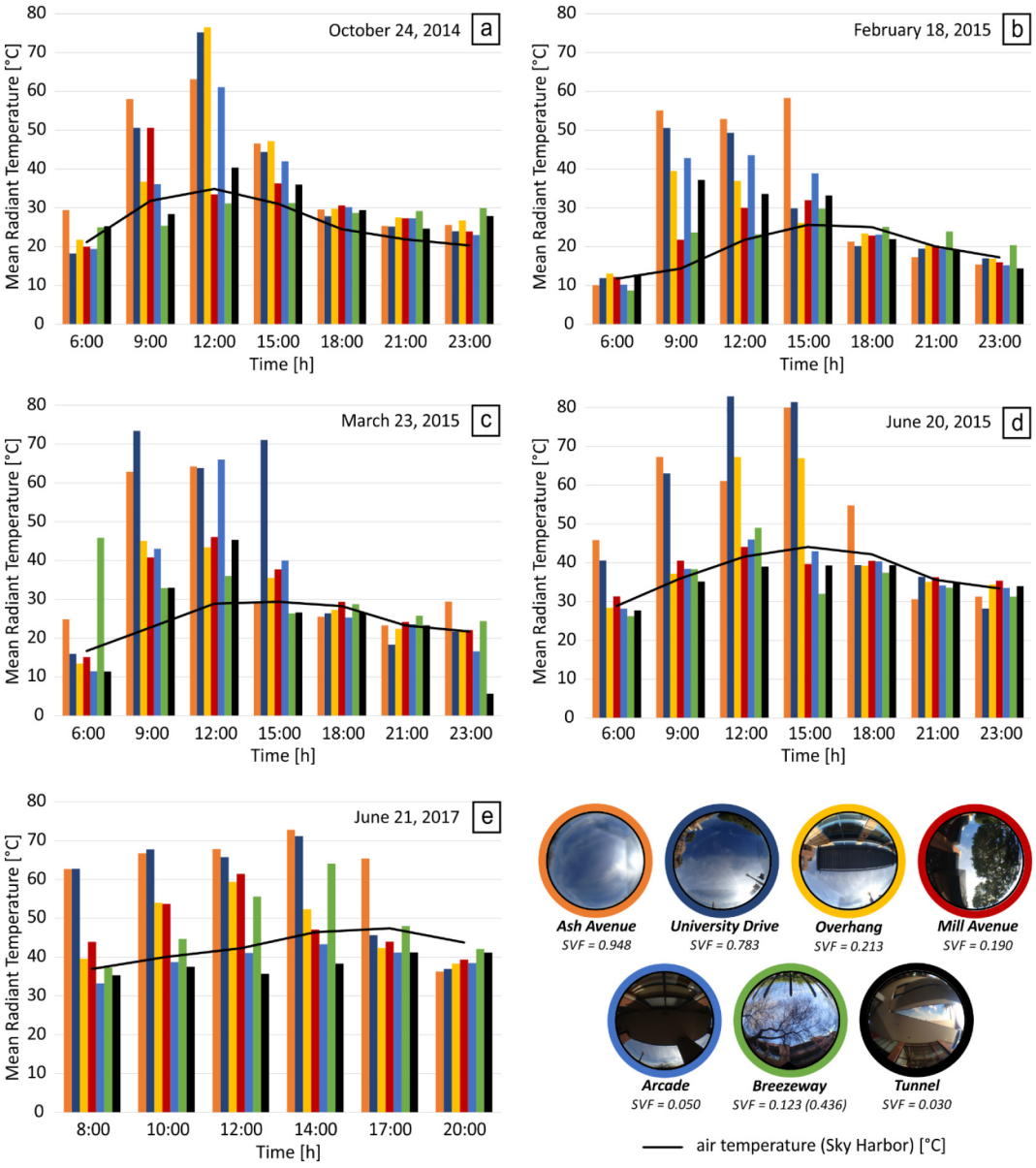


Fig. 3. Mean radiant temperature [°C] observations for each study site and hourly transect by season (a–e) compared to Phoenix Sky Harbor air temperature [°C]; corresponding fisheye views and sky view factors.

The north-south oriented Mill Avenue and east-west oriented Breezeway sites are urban canyons with tree canopy cover. The Breezeway is the most complex location observed in this study with significant horizon limitation changes in the winter due to a deciduous tree to the south. RayMan results are mixed, with the best performance for Mill Avenue (RMSE = 4.3 °C, d = 0.96) and the worst performance for the Breezeway (RMSE = 13.6 °C, d = 0.39) in the spring. For all other seasons, RMSE results are similar to the overall model performance (7.7 °C to 12.4 °C), but the index of agreement is generally low (0.47–0.73). The Breezeway displays a high unsystematic error for the winter (RMSEu = 10.58, RMSEs = 2.85) due to the deciduous trees. While the site is shaded by a definable vegetation canopy in the summer, the horizon is limited by tree branches in the winter that create a random pattern of sun exposure and shade throughout the day. Consistent with other locations, MBE is positive and large in the summer (5.6 °C to 8.2 °C) indicating an overprediction of T_{MRT}.

The east-west overhang is simulated reasonably well for all seasons, with the index of agreement ranging from $d = 0.84$ in the winter to $d = 0.93$ in the summer. The unsystematic error is larger than the systematic error in the winter and shoulder seasons by up to 5.4 °C; the summer exhibits similar unsystematic and systematic errors (RMSEu and RMSEs between 3.9 °C and 4.8 °C). Forcing T_s slightly improves the results.

Observations on the east-west oriented sidewalk shaded by an arcade further confirm the seasonal performance of RayMan. RMSE is lower in the cold season (5.8 °C) and increases with higher T_a (14.3 °C in June 2017). The systematic error follows the same trend and becomes larger as T_a rises (RMSEs = 3.9 °C in February 2015, RMSEs = 13.15 °C in June 2017) while the index of agreement drops from 0.94 to 0.79 (June 2015) and 0.33 (June 2017). Observations under the sheltered walkway confirm RayMan's significant performance drop in simulating T_{MRT} for engineered enclosed spaces with increasing T_a already observed for the tunnel.

3.3. ENVI-met validation

Before evaluating ENVI-met's seasonal T_{MRT} output for the study area, we validate simulated 1.2-m T_a to assess the model's ability to

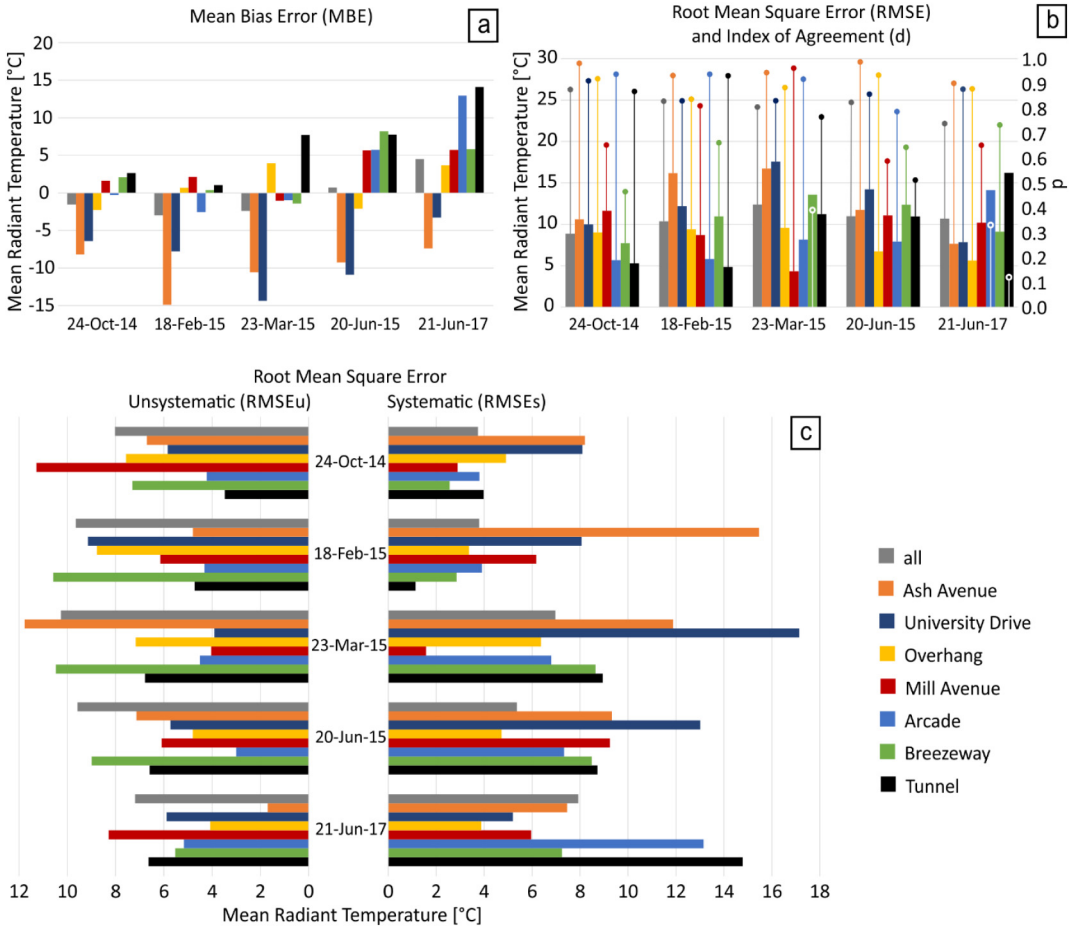


Fig. 4. Mean radiant temperature [°C] statistics for seasonal RayMan simulations: a) Mean Bias Error (MBE); b) Root Mean Square Error RMSE as bars and Index of Agreement (d) as points; c) Unsystematic Root Mean Square Error (RMSEu) and Systematic Root Mean Square Error (RMSEs).

represent physical processes in the urban canopy layer. Seven receptors/observation points (Ash Avenue; University Drive; Overhang; Mill Avenue; Arcade; Breezeway; Tunnel) in the domain allow for a spatially and temporally explicit evaluation (Fig. 5a/b, Supp. Table 4).

Across all seasons, times of day, and receptors (N = 238), RMSE is 2.1 °C with MBE = −0.5 °C and d = 0.99. ENVI-met performs best for June (RMSE is 1.6 °C in 2015 and 1.6 °C in 2017) and worst for February (RMSE is 2.7 °C) with a larger systematic than unsystematic error (RMSEu = 1.3 °C, RMSEs = 2.4 °C). The model overpredicts T_a in the summer (MBE = 0.9 °C), but underpredicts in milder seasons (MBE ranges between −0.9 °C in the fall and −1.8 °C in the spring).

The seasonal index of agreement is good to excellent (0.84 to 1.00) for all locations, but RMSE varies widely across sites. The greatest errors are observed in the winter, with RMSE ranging from 1.7 °C (Breezeway) to 4.2 °C (Ash Avenue) and large systematic errors of up to 3.8 °C (Ash Avenue). The smallest errors occur in June 2017 at the Breezeway (RMSE = 0.8 °C) and University Avenue (RMSE = 1.0 °C).

While the errors are outside the T_a sensor accuracies of 0.5 °C (Kestrel meter) and 0.1 °C (MaRTy), they are within the range of previously reported performance statistics for ENVI-met. Larger error variabilities for individual locations, while optimizing overall fit, suggest that ENVI-met does not accurately model intra-domain T_a distribution in heterogeneous environments, which is probably related to the static diurnal wind profile of the "simple forcing" setting.

Similar to RayMan, ENVI-met exhibits seasonal performance trends in simulating T_{MRT}. For all locations combined, RMSE increases in the hot season (from 11.5 °C in February 2015 to 16.1 °C in June 2017) with negative MBE in the winter and shoulder seasons (−2.4 °C to −0.3 °C) and positive MBE in the summer (6.6 °C to 12.0 °C) (Fig. 5c/d, Fig. 6, Supp. Table 5). The overall index of agreement is satisfactory for all seasons but decreases slightly in the summer (from d = 0.87 in February 2015 to d = 0.73 in June 2017). Site-specific RMSE is higher in June 2017 than 2015 for most locations. At open, unobstructed sites, RMSE is worse than expected with a large unsystematic error, especially at University Avenue in the winter (RMSE = 15.3 °C, RMSEu = 13.2 °C) and Ash Avenue in the spring (RMSE = 18.6 °C, RMSEu = 18.5 °C). March 2015 model output compared with observations show variable performance with inconsistent results across sites. While RMSE is <10.0 °C for Mill Avenue (N-S street canyon), the tunnel, and University Avenue (open), RMSE peaks at 14.5 °C to 18.6 °C for the other sites. Across seasons, there is no detectable pattern that shows one or more sites consistently outperforming others.

In June 2017, all sites that are enclosed by engineered shade (Tunnel, Overhang, Arcade) exhibit large errors with RMSE up to 25.5 °C, RMSEs up to 22.7 °C, and corresponding high positive MBE (12.4 °C to 20.4 °C), indicating that ENVI-met heavily overpredicts T_{MRT} and does not resolve the solar regime correctly at those locations. The Tunnel and Arcade do not yield a significant agreement with observed T_{MRT} as evidenced by low index of agreement values of 0.13 and 0.22, respectively. Additionally, R^2 is 0.001 for the Tunnel, further underlining ENVI-met's inability to model radiation flux densities for enclosed locations under extreme heat conditions.

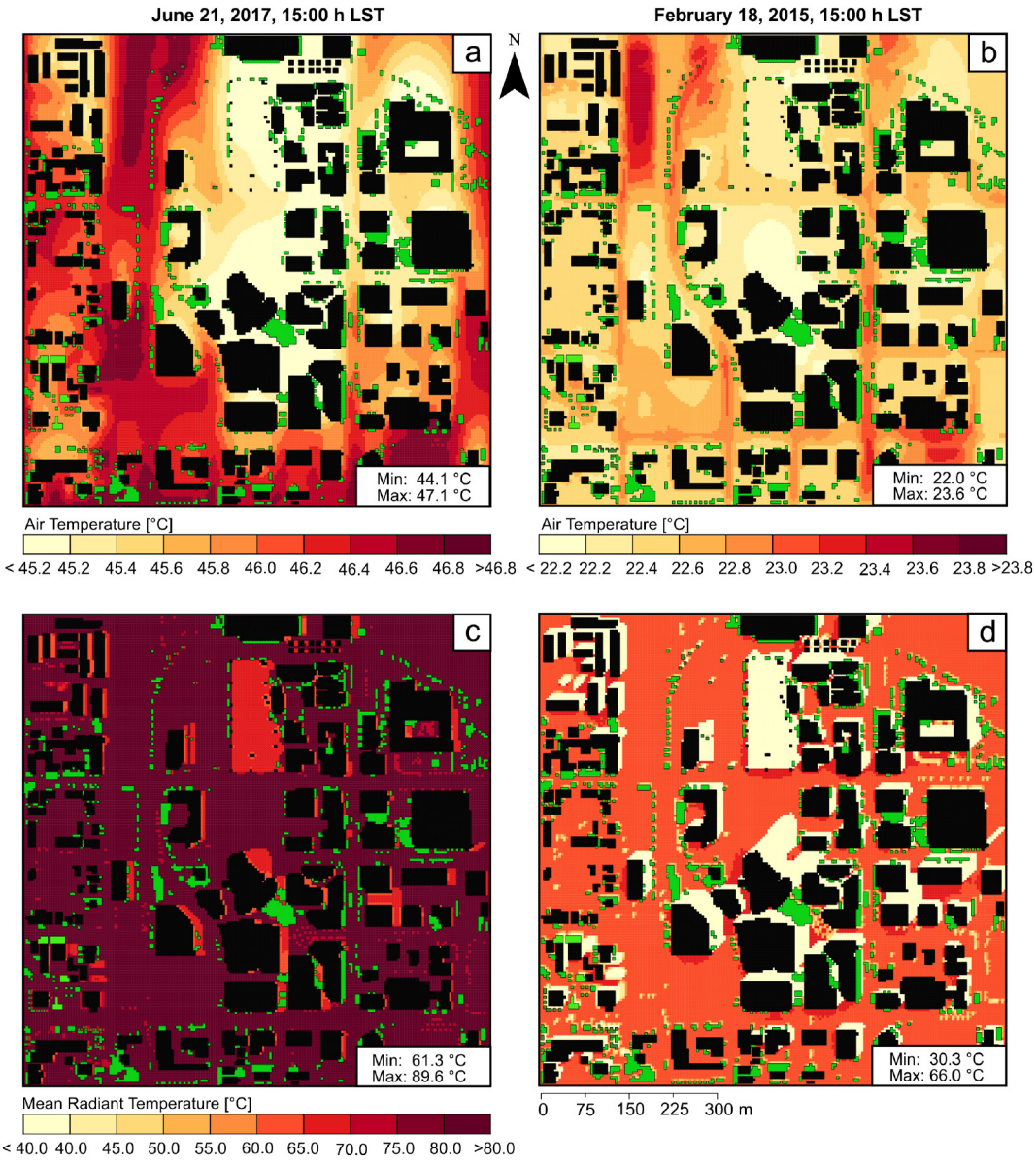


Fig. 5. ENVI-met simulated 1.2-m air temperature at 15:00 h LST for (a) June 21, 2017; (b) February 18, 2015; ENVI-met simulated 1.2-m mean radiant temperature for (c) June 21, 2017; (d) February 18, 2015.

4. Discussion

4.1. Seasonal human-biometeorological field observations

Observed T_{MRT} varies significantly by season, time of day, and location, from a minimum of 8.7 °C (Breezeway, February 19, 2015, 6:00 LST) to a maximum of 84.9 °C (University Drive, June 2015, 12:00 LST). Spatial and temporal T_{MRT} variation is significantly driven by shortwave radiation (Lee et al., 2014; Middel and Krayenhoff, 2019). The differences between T_{MRT} and T_a are most pronounced at 12:00 LST in the summer and 9:00 LST in the winter. While many studies have employed the 6-directional setup to measure radiation flux densities in urban areas for modeling purposes, most observations were conducted in temperate climates (Table 3) with T_a peaking around 29 °C to 35 °C and T_{MRT} in the 40 to 65 °C range (Ali-Toudert and Mayer, 2007; Holst and Mayer, 2011; Kántor et al., 2018; Lee and Mayer, 2018). The wide range of solar radiation regimes in this study allows for a sensitivity analysis and assessment of model performance under cold and extremely hot conditions.

T_{MRT} was observed using two instrument setups, i.e. a Kestrel meter (2014, 2015) and the 6-directional method with MaRTy (2017). Hourly T_{MRT} for June 20, 2015 is slightly higher than June 21, 2017 despite similar sun elevation angles and diurnal T_a profiles. These differences can be attributed to wind conditions and sensor accuracies. While the Kestrel meters have a faster response time than standard 150-mm globe thermometers and allow for stop-and-go transects, they are more sensitive to wind speed due to convective exchange with T_a and lead to an overestimation of T_{MRT} when wind speed is elevated (Teitelbaum et al., 2020). It is also expected that T_{MRT} estimated from Kestrel meter readings is overestimated during mid-day and underestimated in the early morning and evening (Johansson et al., 2014; Thorsson et al., 2007). The difference in sensor performance (Kestrel vs. *MaRTy*) has a direct impact on the validation of the two models. In the summer, the overestimated Kestrel readings likely amplify the positive MBE found across most locations simulated with ENVI-met, further increasing the RMSE. For RayMan, results for open sites would slightly improve using the 6-directional setup. In the shoulder and winter seasons when sun angles are lower, results would be amplified in the opposite direction.

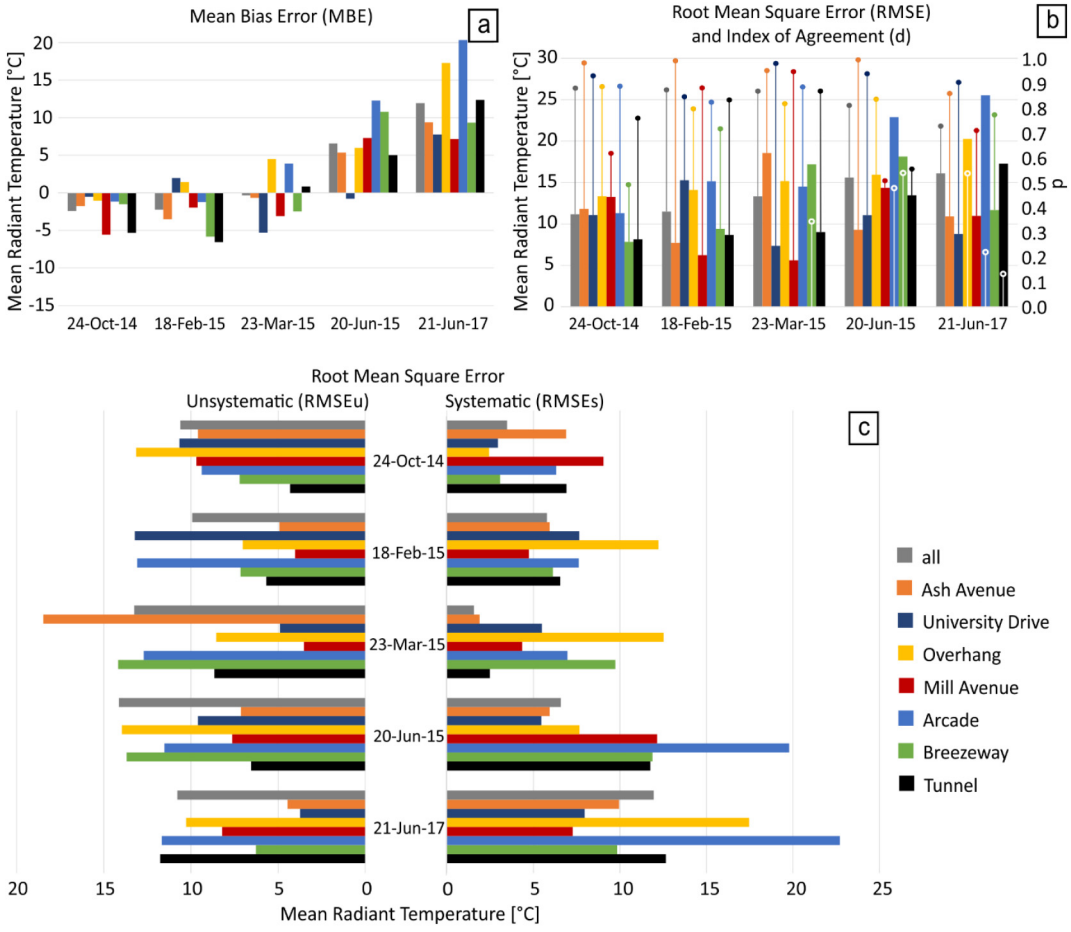


Fig. 6. Mean radiant temperature (T_{MRT}) [°C] statistics for seasonal ENVI-met simulations: a) Mean Bias Error (MBE); b) Root Mean Square Error RMSE as bars and Index of Agreement (d) as points; c) Unsystematic Root Mean Square Error (RMSEu) and Systematic Root Mean Square Error (RMSEs).

In contrast to similar studies validating T_{MRT} in urban areas, our field observations were conducted using transects. Stationary stations have the advantage that the position of the sensors in 3D space remains unchanged during the observation period, while transects allow sampling of a variety of locations using the same sensors. To ensure that the Kestrel meter and *MaRTy* cart were set up at the exact same spot,

Table 3
Other peer-reviewed RayMan and ENVI-met studies that evaluated Mean Radiant Temperature (T_{MRT}) using in-situ observations. Only studies are listed that report the root mean square error (RMSE) of T_{MRT} at a minimum.

Study	Location and validation period	Climate	Model (and grid cell size)	RMSE	Wilmott's *d*
Acero and Arrizabalaga, 2018	Bilbao, Spain (43° N), August 6–8, 2010	Oceanic	ENVI-met v. 4.0, 1.5 × 1.5 × 2, 2 × 2 × 2 (telescoping)	Ta 0.9 °C – 2.1 °C T_{MRT} 7.4 °C – 19.0 °C	0.83–0.94 0.51–0.87
Colter et al., 2019	Phoenix, USA (33ºN), 12 June - 8 July 2015	Arid	RayMan	T_{MRT} 5.5 °C	0.94
Forouzandeh, 2018	Hannover, Germany (52° N), July 7–September 4, 2016; January 27 – February 3, 2017	Temperate	ENVI-met v. 4.1.0, 2 × 2 × 1	Ta 0.7 °C – 3.6 °C T_{MRT} 2.2 °C – 8.4 °C	–
Gál and Kántor, 2020	Szeged, Hungary (46° N), August 7–8, 2016	Temperate	RayMan/ENVI-met v. 4.4.2, 1 × 1 × 0.5	T_{MRT} 8.0 °C/6.9 °C	0.92/0.95
Krüger et al., 2014	Glasgow, UK (55° N), March 3 – July 29, 2011	Temperate	RayMan	T_{MRT} 11.6 °C – 20.2 °C	0.37–0.69
Lee et al., 2016	Freiburg, Germany (48° N), August 4, 2003; July 27, 2009	Temperate	RayMan/ENVI-met v.4 (2013 release), 1 × 1 × 1	T_{MRT} 5.5 °C/12.6 °C	0.95/0.64
Lee and Mayer, 2016	Freiburg, Germany (48° N), July 27, 2009	Temperate	RayMan	T_{MRT} 12.6 °C	0.64
Matzarakis et al., 2010	Freiburg, Germany (48° N), July 16–19, 2006	Temperate	RayMan	T_{MRT} 3.7 °C (open) 0.7 °C (tree)	–
Roth and Lim, 2017	Singapore (1° N), October 1, 8, 15, 2012; January 29, 2013; February 2, 2013; July 21, 24, 28, 2013	Tropical	ENVI-met v. 3.1, 4 × 4 × 4	Ta 0.7 °C – 1.4 °C T_{MRT} 6.4 °C – 14.1 °C (day) T_{MRT} 4.3 °C – 9.2 °C (night)	0.87–0.96 0.77–0.96 0.11–0.86
Zhang et al., 2018	Wuhan, China (31° N), August 16, 2016; January 14, 2017	Subtropical	ENVI-met v. 4.0; 2 × 2 × 0.9 (telescoping)	Ta 1.5 °C (summer), Ta 1.0 °C (winter), T_{MRT} 5.2 °C (summer), T_{MRT} 5.0 °C (winter)	0.91 0.72 0.78 0.76
Zhao and Fong, 2017	Hong Kong, China (22° N), August 5, 2015	Subtropical	ENVI-met v. 4.0, 3 × 3 × 0.6 (telescoping)	Ta 0.6 °C, T_{MRT} 5.7 °C	–
Crank et al., 2020 (this study)	Phoenix, AZ, USA (33° N), October 24, 2014; February 18, 2015; March 23, 2015; June 20, 2015; June 21, 2017	Arid	RayMan/ENVI-met v. 4.3, 4 × 4 × 4	T_{MRT} 8.9 °C – 12.4 °C/Ta 1.5 °C – 2.7 °C, T_{MRT} 11.2 °C – 16.1 °C	0.88/0.99

we marked each location with an X and indicated the orientation of the cart with an arrow to guarantee consistent alignment. The cart was operated by the same person all day, and the Kestrel transects were conducted in pairs of two with at least one of the two people being familiar with the locations. We estimate the spatial offset between transects to be a maximum of 20 cm. Consistency in transect locations across seasons was more difficult, because the chalk markings were not permanent. We took photos of the markings in October 2014 to ensure that the same locations get marked in subsequent seasons and years. We estimate the spatial offset between field campaigns to be less than a meter, which is within the spatial resolution of the established ENVI-met grid and therefore should not have an impact on the model validation.

4.2. RayMan validation

RayMan yields large modeling errors across all field campaigns, with RMSE ranging from 8 °C to 12 °C, irrespective of instrumentation. The magnitude of the error is larger than the T_{MRT} accuracy of ±5 °C suggested by Gál and Kántor (2020) for heat and cold stress studies. Field campaigns in temperate climates found similar accuracy issues that were most pronounced at low sun angles and complex sites. Lee and Mayer (2016) validated RayMan for five heterogeneous sites in Freiburg, Germany for July 27, 2009 and show an overall RMSE of 12.6 °C with large systematic errors and an index of agreement of 0.64. The authors raised concerns about the physical principles implemented in RayMan and recommend updating the governing formulas. Krüger et al. (2014) tested the performance of RayMan in Glasgow during summer 2011 for various input parameter combinations and found that the standard method of running RayMan using fisheye photos, T_a, *RH*, and *v* yielded the largest errors (RMSE = 20.2 °C and $d = 0.37$). They concluded that RayMan is unfit to simulate T_{MRT} without additional input of solar radiation.

Our results confirm previous findings that RayMan does not perform well in complex urban settings with heterogeneous surfaces and horizon limitations, such as the Breezeway location with the deciduous tree that mainly consists of branches during the winter. Errors for tree shaded locations are largely unsystematic, most likely due to the random nature of gaps in the tree crown. RayMan assumes a location is sun-exposed if the fisheye pixel that intersects with the calculated sun position is "white", which can lead to false positives (a location is treated as sun-exposed although it is shaded), especially for low leaf area density trees. Additionally, heterogeneous environments often exhibit highly variable T_s. Park et al. (2018) conducted a sensitivity analysis for T_{MRT} model parameters in MMRT, a multilayer T_{MRT} simulation tool, and found that shade estimates and T_s are most important for predicting T_{MRT}. In our RayMan evaluation, T_{MRT} results improved slightly when adding T_s as an additional forcing parameter, but not significantly. Observed T_s during all field campaigns mostly capture the ground (FLIR) or the lower hemisphere (MaRTy) and are not representative of the average T_s of all surfaces in the 360° surroundings, which requires using special equipment, such as the SMART sensor or thermal panoramas (Aviv et al., n.d.).

Observations at the Tunnel and Arcade sites demonstrate that RayMan is unable to accurately simulate engineered enclosed spaces and consistently overpredicts T_{MRT}, especially in the summer. This further highlights the need to update physical principles of the model to more accurately simulate longwave radiation flux densities, as suggested by Lee and Mayer (2016).

Previously, RayMan has mostly been validated in temperate climates at higher latitudes with a tendency to underestimate T_{MRT}. Matzarakis et al. (2010) report good agreement with 6-directional measurements at two sites for a summer day in Freiburg, Germany with RMSE of 0.7 °C under a tree and 3.7 °C in a semi-open space. Two studies in Taiwan found that RayMan overestimates T_{MRT} by 0.7 °C to 2.3 °C for globe thermometer observed $T_{MRT} = 60$ °C, but did not provide error statistics (Hwang et al., 2011; Lin et al., 2010). Thorsson et al. (2007) simulated T_{MRT} with RayMan for five days in the fall of 2005 and summer of 2006 in Göteborg, Sweden to compare to integral radiation measurements and grey T_g observations. The model underestimated T_{MRT} considerably in the morning and evening for the summer and throughout the whole day in October. The authors point out that reflected shortwave and emitted longwave radiation from surrounding surfaces are important drivers of T_{MRT} at low sun elevation angles that have not been well documented in the RayMan literature. Chen et al. (2014) further corroborate underestimated T_{MRT} values (by about 7 °C) for observations conducted in Freiburg, Germany on August 10, 2010. More recently, Gál and Kántor (2020) evaluated RayMan using integral radiation measurements in downtown Szeged, Hungary for a clear summer day in August 2018 and confirmed that RayMan consistently underestimates T_{MRT} with an RMSE of 8.0 °C.

Our summer results contrast with previous studies due to extreme heat conditions and higher sun elevation angles. While RayMan underestimates T_{MRT} for temperate climate conditions and low sun elevation angles, it overestimates T_{MRT} for hot, dry outdoor conditions when T_a exceeds 40 °C and yields a large positive MBE, especially for semi-open and enclosed sites. Colter et al. (2019) found comparable trends for T_{MRT} measurements under trees in Phoenix parks in 2015 with RMSE and *d* calculated for observed and modeled T_{MRT} of 5.5 °C and 0.94, respectively. Our sensitivity analysis across seasons reveals that RayMan errors become larger with increasing T_a and exceed acceptable levels even for open sites under extreme heat. T_{MRT} errors for Tempe exhibit a clear seasonal trend; values are underestimated in the winter and shoulder seasons and overestimated under hot sunny conditions, doubling the acceptable ±5 °C error for T_{MRT} suggested by Gál and Kántor (2020) for heat stress studies.

4.3. ENVI-met validation

ENVI-met produces large modeling errors across all field campaigns for T_{MRT} (RMSE ranges from 11.2 °C to 16.1 °C) that are outside of the acceptable error range. Yet, similar RMSE values have been reported in the literature (Table 3). Acero and Arrizabalaga (2018) found RMSE ranges of 7.4 °C to 19.0 °C for a neighborhood in Bilboa, Spain; Forouzandeh (2018) conducted ENVI-met simulations for Hannover, Germany that yielded a RMSE of up to 8.4 °C; Lee et al. (2016) reported a RMSE of 12.6 °C for a study site in Freiburg, Germany; and Roth and Lim (2017) found a RMSE of up to 14.1 °C for daytime simulations of a site in tropical Singapore. ENVI-met studies that report an RMSE close to the acceptable error range include two studies in China (Zhang et al., 2018; Zhao and Fong, 2017) with RMSE between 5.0° and 5.7 °C. Despite demonstrated large errors, most studies concluded that the model validation was successful and that ENVI-met is a useful research tool; while some authors caution that it is important to know the limitations of the model, none of the previous studies discouraged its use based on the reported errors.

ENVI-met shows the same seasonal performance trends as RayMan. Compared to winter and shoulder season simulations, RMSE is much larger under extreme heat conditions with positive MBE of up to 12 °C, i.e. the model overestimates T_{MRT}, with a large systematic error of up to 22.7 °C for semi-open and enclosed sites despite good agreement with incoming global solar radiation measurements. The predisposition of the model to overestimate T_{MRT} in hot urban environments has previously been reported by other authors. For example, Elnabawi et al. (2013) simulated an urban canyon in a historic neighborhood of Cairo, Egypt and found that ENVI-met overpredicts T_{MRT} by up to 4 °C in the afternoon. Roth and Lim (2017) evaluated ENVI-met 3.1 for a low-rise neighborhood in Singapore and reported overpredicted T_{MRT} for 6 out of 8 receptor locations. The authors noted that ENVI-met's representation of the parameter f_p, i.e. the proportion of the surface of an upright, rotationally symmetric body that is exposed to direct incoming solar radiation, is different from the standard approach used in human-biometeorological studies. After

correcting for the deviations in f_p, T_{MRT} peak values in the afternoon decreased and daytime magnitudes were mostly lower.

While ENVI-met generally overestimates T_a for extreme heat conditions in this study, which can be attributed to dry soil conditions that disproportionately affect T_a (Roth and Lim, 2017), model performance is surprisingly unsatisfactory for representing intra-neighborhood T_a and T_{MRT} variability across seasons. None of the receptors in the model domain are subject to edge effects, because they are placed away from the model boundaries (Middel et al., 2014). Differences in simulated and observed T_a at receptor locations are most likely related to the diurnal wind regime that is held constant throughout the simulation. For example, the observed wind speed at Ash Avenue was 1.5 m^{-s} at 14:00 LST on June 21, 2017 but simulated as 0.8 m^{-s}. Acero and Arrizabalaga (2018) reported similar discrepancies between modeled and observed wind speed in the domain. To date, no study has systematically assessed the hourly spatial wind profile ENVI-met generates using the "simple forcing" setting to quantify errors in intra-neighborhood T_a variability resulting from the assumption that wind speed and direction are constant throughout the simulation. Although ENVI-met 4.4 offers "full forcing" with variable hourly wind speed and direction, it is unclear if this option produces more accurate results and whether the tradeoff between increased computation time and magnitude of improved accuracy is feasible.

Unsystematic differences in observed and simulated T_{MRT} at semi-enclosed sites can partially be explained through shading errors (Acero and Herranz-Pascual, 2015; Chen et al., 2014; Gál and Kántor, 2020) in the simulated radiation regime. The Arcade in this study is subject to shading errors during midday in the summer: the extent of the horizontal shade structure equals one grid cell (4 m) in ENVI-met, but the Arcade is only about 2.5 m deep. Although the location is shaded all day, ENVI-met simulations for June yield six hours of sun-exposure with added direct shortwave radiation, artificially inflating T_{MRT}. This error is likely due to the coarse vertical grid cell size of 4 m. While ENVI-met can simulate domains with smaller grid cell sizes, this creates a tradeoff between morphological details and performance that could be alleviated by allowing variable grid cell sizes. Choosing our study sites, we tried to avoid locations that were affected by shading errors due to the coarse grid cell size of the ENVI-met domain. The only two locations affected by this error are the arcade and overhang in June, which becomes evident in the large systematic error.

Differences between observed and ENVI-met simulated T_{MRT} for enclosed sites such as the Tunnel exceed 24.0 °C with an RMSE of 17.3 °C in June 2017. This error is almost four times larger than the acceptable error range for heat stress studies. Huttner (2012) outlined limitations to radiative flux calculations in ENVI-met related to diffuse shortwave radiation and vegetation, which were confirmed in an experiment by Gál and Kántor (2020) who found that ENVI-met overestimates T_{MRT} by 9 °C under shaded conditions. Yet, the missing attenuation of diffuse shortwave radiation by trees does not explain the large simulation error at the Tunnel site. Exploration of the directional flux densities at the receptor location shows that the error stems from an overestimation of reflected shortwave radiation.

4.4. Notes on model documentation and validation

Our seasonal T_{MRT} validation for RayMan and ENVI-met demonstrates that both models perform unsatisfactory under extreme heat conditions. Even for open sites, which are less complex due to minimal horizon limitations and limited urban features in the field of view, model errors are much larger than instrumentation errors and fail to meet recommended accuracy requirements for heat stress studies (±5 °C). While systematic and unsystematic errors and model biases can be quantified using in-situ observations, incomplete documentation of the implemented underlying physical principles impedes the detection and elimination of possible errors. This is particularly true for ENVI-met, as the model fully relies on case studies by urban climate community paying customers for validation and model parameter improvement.

Model validation through in-situ observations is vital, not just to improve future model performance but also to assess biases and reliability of current simulation results. This is amplified for software that is not open-sourced or with active community engagement on coding limitations. Even though ENVI-met and RayMan have been widely used by the urban climate and human-biometeorology community in various climatic and morphological contexts for over a decade, many studies did not validate T_{MRT} model results using in-situ observations (e.g., Abdel-Ghany et al., 2014; de Abreu-Harbich et al., 2015; Fröhlich et al., 2019; Ketterer and Matzarakis, 2014; Naboni et al., 2017; Taleghani et al., 2016, 2019). Some ENVI-met studies validated simulated T_a (e.g., Emmanuel and Fernando, 2007; Teshnehdel et al., 2020), which helps assess the model's ability to represent physical processes in domain but does not evaluate radiative flux calculations that drive T_{MRT}. In general, efforts to validate micrometeorological and human-biometeorological variables in ENVI-met are sparse. In a systematic review of ENVI-met studies between 1999 and 2018, Tsoka et al. (2018) reported 52 evaluation studies out of almost 280 reviewed, and only 16 studies validated T_{MRT}. Similarly, a Scopus search for "RayMan" in June 2020 yielded 87 studies that used the model between 2007 and 2020, but only 11 studies evaluated T_{MRT}. Without evaluating the model output being used in analysis, there is insufficient evidence from the results from any model to make conclusions or provide policy recommendations to stakeholders.

While RayMan and ENVI-met are the most popular models to date for radiation flux modeling and thermal comfort assessments, other models have been developed that could not be assessed within the limited scope of this study. SOLWEIG (Lindberg et al., 2008; Lindberg and Grimmond, 2011) was specifically developed to simulate spatial and temporal variations of 3D radiant fluxes and T_{MRT} in complex urban environments. Its performance in comparison to ENVI-met and RayMan has been assessed by various authors (Chen et al., 2014; Gál and Kántor, 2020; Liu et al., 2020). VTUF-3D 1.0 (Nice et al., 2018) builds on the urban micro-climate surface energy balance model TUF-3D (Krayenhoff and Voogt, 2007) and provides estimates of T_s, T_{MRT}, and the thermal comfort index UTCI. Park et al. (2018) developed a multilayer T_{MRT} model (MMRT) that calculates the radiation transfer in complex environments using Monte Carlo Ray Tracing. Another promising approach from the architecture community to calculate T_{MRT} is the Ladybug/Honeybee plugin for Grasshopper (Evola et al., 2020; Lobaccaro et al., 2018; Naboni et al., 2017; Roudsari and Pak, 2013) and Cohen et al. (2020) developed Mr.T, an EXCEL spreadsheet based software that calculates T_{MRT} for individual locations, analogous to RayMan. ENVI-met counterparts include OpenFOAM, Star-CCM+, and FLUENT; however, not all have been tested or validated for T_{MRT} (Botham-Myint et al., 2015). Future research should conduct a systematic comparison across various T_{MRT} models using high quality 6-directional flux density measurements in various climatic zones and urban contexts across seasons as benchmarks to perform sensitivity analyses and a full assessment of model performance to guide future model development.

5. Conclusions

This study validated RayMan and ENVI-met T_{MRT} simulations for Tempe, Arizona's Mill Avenue district using in-situ observations with Kestrel meters and MaRTy to assess model performance across seasons and under extreme heat conditions. Observations covered a wide range of solar radiation regimes from a minimum observed T_{MRT} of 8.7 °C during winter to a maximum of 84.9 °C in the summer. Both models produced large errors across seasons irrespective of instrumentation (Kestrel vs. MaRTy), with RMSE ranging from 8 °C to 12 °C (RayMan) and 11.2 °C to 16.1 °C (ENVI-met), exceeding a suggested T_{MRT} accuracy of ±5 °C for heat and cold stress studies.

Our results show that RayMan does not work well for engineered enclosed spaces (e.g. tunnel, arcade), complex settings (e.g., low leaf area density trees or deciduous trees during leaf-off season), and under extreme heat conditions. ENVI-met exhibits the same seasonal

swing to RMSE and MBE as RayMan, underestimating T_{MRT} in colder seasons and overestimating in the summer while errors become larger with increasing T_a towards extreme heat (RMSE up to 25.5 °C). ENVI-met did not resolve intra-domain spatial variability of T_a and T_{MRT}. This can be partially attributed to constant diurnal wind forcing (T_a) and the uniformly gridded nature of ENVI-met that creates a tradeoff between detailed urban morphology representation and model performance (T_{MRT}), leading to errors in the modeled radiation regime. For enclosed spaces, ENVI-met estimates reflected shortwave radiation to be equal to sun-exposed locations, which is likely a bug in the code that produces highly inaccurate results for T_{MRT} under engineered shade, e.g. from shade sails.

Based on our model assessment, we do not recommend using ENVI-met and RayMan for thermal comfort, exposure, and shade studies in hot environments without extensive in-situ T_{MRT} observations for bias evaluation, as both models struggle with radiation flux calculations of enclosed spaces, and errors get larger as incoming shortwave radiation increases. More generally, we caution against using models in micrometeorological or morphological extremes that they were not designed for without in-situ validation to assess model performance. While models may yield satisfactory results for temperate climates, extreme conditions could amplify errors inherent in the radiative flux modeling. Model validation should be performed across multiple points in space, time, urban forms, and, ideally, include the full parameter space of output variables, but at a minimum, the variable of interest (Krayenhoff et al., n.d.). In interpreting results, users should be aware of model limitations due to implemented physical principles, parameterizations, and assumptions. Rigorous validation will advance model development and improve our understanding of the complex dynamics in the urban canopy layer through observations and modeling.

CRediT authorship contribution statement

Peter J. Crank: Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Ariane Middel:** Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review & editing. **Melissa Wagner:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Dani Hoots:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Martin Smith:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Anthony Brazel:** Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2020.141392.

References

Abdel-Ghany, A.M., Al-Helal, I.M., Shady, M.R., 2014. Evaluation of human thermal comfort and heat stress in an outdoor urban setting in summer under arid climatic conditions. Environ. Prot. Eng. 40, 139–150. https://doi.org/10.5277/epel40311.

de Abreu-Harbich, L.V., Labaki, L.C., Matzarakis, A., 2015. Effect of tree planting design and tree species on human thermal comfort in the tropics. Landsc. Urban Plan. 138, 99–109. https://doi.org/10.1016/j.landurbplan.2015.02.008.

Acero, J.A., Arrizabalaga, J., 2018. Evaluating the performance of ENVI-met model in diurnal cycles for different meteorological conditions. Theor. Appl. Climatol. 131, 455–469. https://doi.org/10.1007/s00704-016-1971-y.

Acero, J.A., Herranz-Pascual, K., 2015. A comparison of thermal comfort conditions in four urban spaces by means of measurements and modelling techniques. Build. Environ. 93, 245–257. https://doi.org/10.1016/j.buildenv.2015.06.028.

Ali-Toudert, F., Mayer, H., 2007. Thermal comfort in an east–west oriented street canyon in Freiburg (Germany) under hot summer conditions. Theor. Appl. Climatol. 87, 223–237. https://doi.org/10.1007/s00704-005-0194-4.

Aminipouri, M., Knudby, A.J., Krayenhoff, E.S., Zickfeld, K., Middel, A., 2019a. Modelling the impact of increased street tree cover on mean radiant temperature across Vancouver's local climate zones. Urban Forestry and Urban Greening 39, 9–17. https://doi.org/10.1016/j.ufug.2019.01.016.

Aminipouri, M., Rayner, D., Lindberg, F., Thorsson, S., Knudby, A.J., Zickfeld, K., Middel, A., Krayenhoff, E.S., 2019b. Urban tree planting to maintain outdoor thermal comfort under climate change: the case of Vancouver's local climate zones. Build. Environ. 158, 226–236. https://doi.org/10.1016/j.buildenv.2019.05.022.

Aviv, D., Guo, H., Middel, A., Meggers, F., n.d. Evaluating radiant heat in an outdoor urban environment: resolving spatial and temporal variations with two sensing platforms and data-driven simulation. Urban Clim.

Botham-Myint, D., Recktenwald, G.W., Sailor, D.J., 2015. Thermal footprint effect of rooftop urban cooling strategies. Urban Clim. 14, 268–277. https://doi.org/10.1016/j.uclim.2015.07.005.

Bröde, P., Fiala, D., B\la\.zejczyk, K., Holmér, I., Jendritzky, G., Kampmann, B., Tinz, B., Havenith, G., 2012. Deriving the operational procedure for the Universal Thermal Climate Index (UTCI). Int. J. Biometeorol. 56, 481–494. https://doi.org/10.1007/s00484-011-0454-1.

Bruse, M., Fleer, H., 1998. Simulating surface-plant-air interactions inside urban environments with a three dimensional numerical model. Environ. Model. Softw. 13, 373–384. https://doi.org/10.1016/S1364-8152(98)00042-5.

Chen, L., Ng, E., 2012. Outdoor thermal comfort and outdoor activities: a review of research in the past decade. Cities 29, 118–125. https://doi.org/10.1016/j.cities.2011.08.006.

Chen, Y.C., Lin, T.P., Matzarakis, A., 2014. Comparison of mean radiant temperature from field experiment and modelling: a case study in Freiburg, Germany. Theor. Appl. Climatol. 118, 535–551. https://doi.org/10.1007/s00704-013-1081-z.

Chow, W.T.L., Brazel, A.J., 2012. Assessing xeriscaping as a sustainable heat island mitigation approach for a desert city. Build. Environ. 47, 170–181. https://doi.org/10.1016/j.buildenv.2011.07.027.

Cohen, S., Palatchi, Y., Palatchi, D.P., Shashua-Bar, L., Lukyanov, V., Yaakov, Y., Matzarakis, A., Tanny, J., Potchter, O., 2020. Mean radiant temperature in urban canyons from solar calculations, climate and surface properties – theory, validation and 'Mr.T' software. Build. Environ., 106927 https://doi.org/10.1016/j.buildenv.2020.106927.

Colter, K.R., Middel, A.C., Martin, C.A., 2019. Effects of natural and artificial shade on human thermal comfort in residential neighborhood parks of Phoenix, Arizona, USA. Urban Forestry and Urban Greening 44. https://doi.org/10.1016/j.ufug.2019.126429.

Crank, P.J., Sailor, D.J., Ban-Weiss, G., Taleghani, M., 2018. Evaluating the ENVI-met microscale model for suitability in analysis of targeted urban heat mitigation strategies. Urban Clim. 26, 188–197.

Crewe, K., Brazel, A., Middel, A., 2016. Desert new urbanism: testing for comfort in downtown Tempe, Arizona. J. Urban Des. 21, 746–763. https://doi.org/10.1080/13574809.2016.1187558.

Elnabawi, M.H., Hamza, N., Dudek, S., 2013. Use and evaluation of the envi-met model for two different urban forms in Cairo, Egypt: measurements and model simulations. Proceedings of BS 2013: 13th Conference of the International Building Performance Simulation Association, pp. 2800–2806.

Emmanuel, R., Fernando, H.J.S., 2007. Urban heat islands in humid and arid climates: role of urban form and thermal properties in Colombo, Sri Lanka and Phoenix, USA. Clim. Res. 34, 241–251. https://doi.org/10.3354/cr00694.

Emmanuel, R., Rosenlund, H., Johansson, E., 2007. Urban shading—a design option for the tropics? A study in Colombo, Sri Lanka. Int. J. Climatol. 27, 1995–2004. https://doi.org/10.1002/joc.1609.

Evola, G., Costanzo, V., Magrì, C., Margani, G., Marletta, L., Naboni, E., 2020. A novel comprehensive workflow for modelling outdoor thermal comfort and energy demand in urban canyons: results and critical issues. Energy & Buildings 216, 109946. https://doi.org/10.1016/j.enbuild.2020.109946.

Forouzandeh, A., 2018. Numerical modeling validation for the microclimate thermal condition of semi-closed courtyard spaces between buildings. Sustain. Cities Soc. 36, 327–345. https://doi.org/10.1016/j.scs.2017.07.025.

Fröhlich, D., Gangwisch, M., Matzarakis, A., 2019. Effect of radiation and wind on thermal comfort in urban environments - application of the ray man and sky Helios model. Urban Clim. 27, 1–7. https://doi.org/10.1016/j.uclim.2018.10.006.

Gál, C.V., Kántor, N., 2020. Modeling mean radiant temperature in outdoor spaces, a comparative numerical simulation and validation study. Urban Clim. 32, 100571. https://doi.org/10.1016/j.uclim.2019.100571.

Häb, K., Feige, N.H., Huettenberger, L.S., Middel, A., Hagen, H., 2014. Visualizing the temporal development of thermo-radiative features on ground-based thermographs. Environ. Earth Sci. 72, 3781–3793. https://doi.org/10.1007/s12665-014-3472-6.

Häb, K., Ruddell, B.L., Middel, A., 2015. Sensor lag correction for mobile urban microclimate measurements. Urban Clim. 14, 622–635. https://doi.org/10.1016/j.uclim.2015.10.003.

Hedquist, B.C., Brazel, A.J., 2014. Seasonal variability of temperatures and outdoor human comfort in Phoenix, Arizona, U.S.A. Build. Environ. 72, 377–388. https://doi.org/10.1016/j.buildenv.2013.11.018.

Heris, M.P., Middel, A., Muller, B., 2020. Impacts of form and design policies on urban microclimate: assessment of zoning and design guideline choices in urban redevelopment projects. Landsc. Urban Plan. 202, 103870. https://doi.org/10.1016/j.landurbplan.2020.103870.

Holst, J., Mayer, H., 2011. Impacts of street design parameters on human-biometeorological variables. Meteorol. Z. 20, 541–552. https://doi.org/10.1127/0941-2948/2011/0254.

Höppe, P., 1992. A new procedure to determine the mean radiant temperature outdoors. Wetter und Leben 44, 147–151.

Huttner, S., 2012. Further Development and Application of the 3D Microclimate Simulation ENVI-Met. Dissertation. Johannes Gutenberg-University of Mainz, Germany http://ubm.opus.hbznrw.de/volltexte/2012/3112/pdf/doc.pdf. (Accessed 7 July 2020).
Hwang, R.L., Lin, T.P., Matzarakis, A., 2011. Seasonal effects of urban street shading on long-term outdoor thermal comfort. Build. Environ. 46, 863–870. https://doi.org/10.1016/j.buildenv.2010.10.017.
ISO, 1998. ISO Standard 7726, Ergonomics of the Thermal Environment, Instruments for Measuring Physical Quantities. ISO, Geneva.
Johansson, E., Thorsson, S., Emmanuel, R., Krüger, E., 2014. Instruments and methods in outdoor thermal comfort studies - the need for standardization. Urban Clim. 10, 346–366. https://doi.org/10.1016/j.uclim.2013.12.002.
Jones, B., O'Neill, B.C., Mcdaniel, L., Mcginnis, S., Mearns, L.O., Tebaldi, C., 2015. Future population exposure to US heat extremes. Nat. Clim. Chang. 5, 652–655. https://doi.org/10.1038/nclimate2631.
Kántor, N., Chen, L., Gál, C.V., 2018. Human-biometeorological significance of shading in urban public spaces—summertime measurements in Pécs, Hungary. Landsc. Urban Plan. 170, 241–255. https://doi.org/10.1016/j.landurbplan.2017.09.030.
Ketterer, C., Matzarakis, A., 2014. Human-biometeorological assessment of heat stress reduction by replanning measures in Stuttgart, Germany. Landsc. Urban Plan. 122, 78–88. https://doi.org/10.1016/j.landurbplan.2013.11.003.
Krayenhoff, E.S., Voogt, J.A., 2007. A microscale three-dimensional urban energy balance model for studying surface temperatures. Boundary-Layer Meteorol 123, 433–461. https://doi.org/10.1007/s10546-006-9153-6.
Krayenhoff, S.E., Broadbent, A.M., Zhao, L., Georgescu, M., Middel, A., Voogt, J.A., Martilli, A., Sailor, D., Erell, E., n.d. Cooling hot cities: a systematic and critical review of the numerical modelling literature. Environ. Res. Lett.
Krüger, E.L., Minella, F.O., Matzarakis, A., 2014. Comparison of different methods of estimating the mean radiant temperature in outdoor thermal comfort studies. Int. J. Biometeorol. 58, 1727–1737. https://doi.org/10.1007/s00484-013-0777-1.
Lee, H., Mayer, H., 2016. Validation of the mean radiant temperature simulated by the Ray Man software in urban environments. Int. J. Biometeorol. 60, 1775–1785. https://doi.org/10.1007/s00484-016-1166-3.
Lee, H., Mayer, H., 2018. Maximum extent of human heat stress reduction on building areas due to urban greening. Urban Forestry and Urban Greening 32, 154–167. https://doi.org/10.1016/j.ufug.2018.04.010.
Lee, H., Holst, J., Mayer, H., 2013. Modification of human-biometeorologically significant radiant flux densities by shading as local method to mitigate heat stress in summer within urban street canyons. Adv. Meteorol. 2013. https://doi.org/10.1155/2013/312572.
Lee, H., Mayer, H., Schindler, D., 2014. Importance of 3-D radiant flux densities for outdoor human thermal comfort on clear-sky summer days in Freiburg, Southwest Germany. Meteorol. Z., 315–330 https://doi.org/10.1127/0941-2948/2014/0536.
Lee, H., Mayer, H., Chen, L., 2016. Contribution of trees and grasslands to the mitigation of human heat stress in a residential district of Freiburg, Southwest Germany. Landsc. Urban Plan. 148, 37–50. https://doi.org/10.1016/j.landurbplan.2015.12.004.
Lin, T.P., Matzarakis, A., 2008. Tourism climate and thermal comfort in Sun Moon Lake, Taiwan. Int. J. Biometeorol. 52, 281–290. https://doi.org/10.1007/s00484-007-0122-7.
Lin, T.P., Matzarakis, A., Hwang, R.L., 2010. Shading effect on long-term outdoor thermal comfort. Build. Environ. 45, 213–221. https://doi.org/10.1016/j.buildenv.2009.06.002.
Lindberg, F., Grimmond, C.S.B., 2011. Nature of vegetation and building morphology characteristics across a city: influence on shadow patterns and mean radiant temperatures in London. Urban Ecosyst. 14, 617–634. https://doi.org/10.1007/s11252-011-0184-5.
Lindberg, F., Holmer, B., Thorsson, S., 2008. SOLWEIG 1.0- Modelling spatial variations of 3D radiant fluxes and mean radiant temperature in complex urban settings. Int. J. Biometeorol. 52, 697–713. https://doi.org/10.1007/s00484-008-0162-7.
Liu, D., Hu, S., Liu, J., 2020. Contrasting the performance capabilities of urban radiation field between three microclimate simulation tools. Build. Environ. 175. https://doi.org/10.1016/j.buildenv.2020.106789.
Lobaccaro, G., Croce, S., Vettorato, D., Carlucci, S., 2018. A holistic approach to assess the exploitation of renewable energy sources for design interventions in the early design phases. Energy & Buildings 175, 235–256. https://doi.org/10.1016/j.enbuild.2018.06.066.
Mahmoud, A.H.A., 2011. Analysis of the microclimatic and human comfort conditions in an urban park in hot and arid regions. Build. Environ. 46, 2641–2656. https://doi.org/10.1016/j.buildenv.2011.06.025.
Matzarakis, A., Rutz, F., Mayer, H., 2007. Modelling radiation fluxes in simple and complex environments - application of the Ray Man model. Int. J. Biometeorol. 51, 323–334. https://doi.org/10.1007/s00484-006-0061-8.
Matzarakis, A., Rutz, F., Mayer, H., 2010. Modelling radiation fluxes in simple and complex environments: basics of the Ray Man model. Int. J. Biometeorol. 54, 131–139. https://doi.org/10.1007/s00484-009-0261-0.
Mayer, H., Höppe, P., 1987. Thermal comfort of man in different urban environments. Theor. Appl. Climatol. 38, 43–49. https://doi.org/10.1007/BF00866252.
Mayer, H., Holst, J., Dostal, P., Imbery, F., Schindler, D., 2008. Human thermal comfort in summer within an urban street canyon in Central Europe. Meteorol. Z., 241–250 https://doi.org/10.1127/0941-2948/2008/0285.
Middel, A., Krayenhoff, E.S., 2019. Micrometeorological determinants of pedestrian thermal exposure during record-breaking heat in Tempe, Arizona: introducing the MaRTy observational platform. Sci. Total Environ. https://doi.org/10.1016/j.scitotenv.2019.06.085.
Middel, A., Hab, K., Brazel, A.J., Martin, C.A., Guhathakurta, S., 2014. Impact of urban form and design on mid-afternoon microclimate in Phoenix local climate zones. Landsc. Urban Plan. 122, 16–28. https://doi.org/10.1016/j.landurbplan.2013.11.004.
Middel, A., Chhetri, N., Quay, R., 2015. Urban forestry and cool roofs: assessment of heat mitigation strategies in Phoenix residential neighborhoods. Urban Forestry and Urban Greening 14, 178–186. https://doi.org/10.1016/j.ufug.2014.09.010.
Middel, A., Selover, N., Hagen, B., Chhetri, N., 2016. Impact of shade on outdoor thermal comfort—a seasonal field study in Tempe, Arizona. Int. J. Biometeorol. 60, 1849–1861. https://doi.org/10.1007/s00484-016-1172-5.
Middel, A., Lukasczyk, J., Maciejewski, R., Demuzere, M., Roth, M., 2018. Sky view factor footprints for urban climate modeling. Urban Clim. 25, 120–134. https://doi.org/10.1016/j.uclim.2018.05.004.
Middel, A., Lukasczyk, J., Zakrzewski, S., Arnold, M., Maciejewski, R., 2019. Urban form and composition of street canyons: a human-centric big data and deep learning approach. Landsc. Urban Plan. 183, 122–132. https://doi.org/10.1016/j.landurbplan.2018.12.001.
Middel, A., Turner, V.K., Schneider, F.A., Zhang, Y., Stiller, M., 2020. Solar reflective pavement – a policy panacea to heat mitigation? Environ. Res. Lett. 15 (6), 064016. https://doi.org/10.1088/1748-9326/ab87d4.
Morakinyo, T.E., Kong, L., Lau, K.K.L., Yuan, C., Ng, E., 2017. A study on the impact of shadow-cast and tree species on in-canyon and neighborhood's thermal comfort. Build. Environ. 115, 1–17. https://doi.org/10.1016/j.buildenv.2017.01.005.
Naboni, E., Meloni, M., Coccolo, S., Kaempf, J., Scartezzini, J.L., 2017. An overview of simulation tools for predicting the mean radiant temperature in an outdoor space. Energy Procedia, 1111–1116 https://doi.org/10.1016/j.egypro.2017.07.471.
Nice, K.A., Coutts, A.M., Tapper, N.J., 2018. Development of the VTUF-3D v1.0 urban micro-climate model to support assessment of urban vegetation influences on human thermal comfort. Urban Clim. 24, 1052–1076. https://doi.org/10.1016/j.uclim.2017.12.008.
Park, C.Y., Lee, D.K., Krayenhoff, E.S., Heo, H.K., Ahn, S., Asawa, T., Murakami, A., Kim, H.G., 2018. A multilayer mean radiant temperature model for pedestrians in a street canyon with trees. Build. Environ. 141, 298–309. https://doi.org/10.1016/j.buildenv.2018.05.058.
Roth, M., Lim, V.H., 2017. Evaluation of canopy-layer air and mean radiant temperature simulations by a microclimate model over a tropical residential neighbourhood. Build. Environ. 112, 177–189. https://doi.org/10.1016/j.buildenv.2016.11.026.
Roudsari, M.S., Pak, M., 2013. Ladybug: a parametric environmental plugin for grasshopper to help designers create an environmentally-conscious design. Proceedings of BS 2013: 13th Conference of the International Building Performance Simulation Association.
Stewart, I.D., Oke, T.R., 2012. Local climate zones for urban temperature studies. Bull. Am. Meteorol. Soc. 93, 1879–1900. https://doi.org/10.1175/BAMS-D-11-00019.1.
Taleghani, M., Sailor, D., Ban-Weiss, G.A., 2016. Micrometeorological simulations to predict the impacts of heat mitigation strategies on pedestrian thermal comfort in a Los Angeles neighborhood. Environ. Res. Lett. 11, 024003. https://doi.org/10.1088/1748-9326/11/2/024003.
Taleghani, M., Crank, P.J., Mohegh, A., Sailor, D.J., Ban-Weiss, G.A., 2019. The impact of heat mitigation strategies on the energy balance of a neighborhood in Los Angeles. Sol. Energy 177, 604–611.
Teitelbaum, E., Chen, K.W., Meggers, F., Guo, H., Houchois, N., Pantelic, J., Rysanek, A., 2020. Globe thermometer free convection error potentials. Sci. Rep. 10. https://doi.org/10.1038/s41598-020-59441-1.
Teshnehdel, S., Akbari, H., Di Giuseppe, E., Brown, R.D., 2020. Effect of tree cover and tree species on microclimate and pedestrian comfort in a residential district in Iran. Build. Environ. 178. https://doi.org/10.1016/j.buildenv.2020.106899.
Thorsson, S., Lindberg, F., Eliasson, I., Holmer, B., 2007. Different methods for estimating the mean radiant temperature in an outdoor urban setting. Int. J. Climatol. 27, 1983–1993. https://doi.org/10.1002/joc.1537.
Thorsson, S., Rockl??v, J., Konarska, J., Lindberg, F., Holmer, B., Dousset, B., Rayner, D., 2014. Mean radiant temperature - a predictor of heat related mortality. Urban Clim. 10, 332–345. https://doi.org/10.1016/j.uclim.2014.01.004.
Tsoka, S., Tsikaloudaki, A., Theodosiou, T., 2018. Analyzing the ENVI-Met Microclimate Model's Performance and Assessing Cool Materials and Urban Vegetation Applications-A Review. https://doi.org/10.1016/j.scs.2018.08.009.
UN, 2019. World Urbanization Prospects: The 2018 Revision, World Urbanization Prospects: The 2018 Revision. https://doi.org/10.18356/b9e995fe-en.
Western Regional Climate Center, 2020. Historical Climate Information. URL https://wrcc.dri.edu/cgi-bin/cliMAIN.pl?az8499 (accessed July 7, 2020).
Willmott, C.J., 1981. On the validation of models. Phys. Geogr. 2 (2), 184–194. https://doi.org/10.1080/02723646.1981.10642213.
Yahia, M.W., Johansson, E., 2013. Influence of urban planning regulations on the microclimate in a hot dry climate: the example of Damascus, Syria. J. Housing Built Environ. 28, 51–65. https://doi.org/10.1007/s10901-012-9280-y.
Zhang, L., Zhan, Q., Lan, Y., 2018. Effects of the tree distribution and species on outdoor environment conditions in a hot summer and cold winter zone: a case study in Wuhan residential quarters. Build. Environ. 130, 27–39. https://doi.org/10.1016/j.buildenv.2017.12.014.
Zhao, T.F., Fong, K.F., 2017. Characterization of different heat mitigation strategies in landscape to fight against heat island and improve thermal comfort in hot-humid climate (part II): evaluation and characterization. Sustain. Cities Soc. 35, 841–850. https://doi.org/10.1016/j.scs.2017.05.006.
Zölch, T., Maderspacher, J., Wamsler, C., Pauleit, S., 2016. Using green infrastructure for urban climate-proofing: an evaluation of heat mitigation measures at the micro-scale. Urban Forestry and Urban Greening 20, 305–316. https://doi.org/10.1016/j.ufug.2016.09.011.